

The Singular Rotating-Wave Cutoff Limit Converges in Norm Resolvent Sense

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Status: literature already answered (affirmative under the requested additional structure; no new theorem is claimed here).

The 2022 question

Lonigro constructs singular rotating-wave spin-boson Hamiltonians with massive dispersion and form factor $f \in \mathcal{H}_{-s}^r \subset \mathcal{H}_{-s}$, $1 < s \leq 2$ and $s - 1 \leq r \leq 1$. After Theorem 2, he notes that the genuinely renormalized cases are obtained from regular cutoffs only in strong resolvent sense and asks whether this can be improved, at least under additional assumptions [1] (source PDF page 10).

Direct later answer

Hinrichs, Lampart, and Valentín Martín treat

$$H_{\text{SB},\Lambda} = A \otimes 1 + 1 \otimes d\Gamma(\omega) + B^* \otimes a(v_\Lambda) + B \otimes a^*(v_\Lambda).$$

Their Example 2 identifies the rotating-wave approximation by $B = \sigma_-$, hence $B^2 = 0$ [2] (supporting PDF page 2). Their Theorem 2.5 (supporting PDF page 6) proves norm-resolvent convergence of the renormalized regular approximants whenever the normal component is absent. Consequently, for the pure rotating-wave interaction and $(1 + \omega)^{-1}v \in L^2$,

$$H_{\text{SB},\Lambda} + B^*BE_\Lambda \xrightarrow[\Lambda \rightarrow \infty]{\text{norm resolvent}} H_{\text{SB}}, \quad E_\Lambda = \|\omega^{-1/2}v_\Lambda\|_2^2.$$

Here B^*B is the excited-state projection. Thus this operator counterterm is exactly a cutoff-dependent bare excitation-energy shift, matching the source renormalization (with the coupling absorbed into v).

The source is massive. On that domain, $f \in \mathcal{H}_{-s}^r \subset \mathcal{H}_{-s}$ for $1 < s \leq 2$ implies the later critical condition, and spectral cutoffs converge in its weighted b_2 norm. Hence the theorem covers the source's entire singular range, including $s = 2$. The later authors explicitly state that their result extends Lonigro's rotating-wave construction to arbitrary coupling, massless fields, and norm-resolvent convergence.

Idea of the later proof

For a 2-nilpotent interaction, the interior-boundary-condition construction factorizes the Hamiltonian through

$$(1 + G)(d\Gamma(\omega) + \lambda - T)(1 + G^*), \quad G^2 = 0,$$

so $(1 + G)^{-1} = 1 - G$. The weighted b_2 topology makes G and the renormalized self-energy T norm-continuous under cutoff removal. Proposition 3.5 converts this into norm-resolvent continuity of the factorized operator; Theorem 2.5 then adds the bounded spin term. The nilpotent case avoids the merely strong continuity of a nontrivial dressing unitary, which is why the endpoint still has norm-resolvent convergence.

Conclusion and scope

The open improvement question therefore has an affirmative answer for the same rotating-wave model under its intrinsic 2-nilpotent structure, and in fact with broader infrared and coupling scope. This packet does not claim that arbitrary nonnormal, nonnilpotent spin couplings enjoy the same limit.

References

- [1] D. Lonigro, *Renormalization of spin-boson interactions mediated by singular form factors*, arXiv:2210.15267 (2022).
- [2] B. Hinrichs, J. Lampart, and J. Valentín Martín, *Ultraviolet Renormalization of Spin Boson Models I. Normal and 2-Nilpotent Interactions*, arXiv:2502.04876 (2025).